

Amazon EKS Hands-On Learning Journey

Overview

This repository contains the YAML manifests, configurations, and practical exercises used during a complete hands-on Amazon EKS (Elastic Kubernetes Service) learning journey.

The goal of this project was not just to learn Kubernetes concepts theoretically, but to deploy, operate, troubleshoot, monitor, and eventually clean up a real EKS environment on AWS.

By the end of this journey, the cluster was built from scratch, applications were deployed, storage was configured, workloads were scaled, monitoring was enabled, and the entire infrastructure was safely decommissioned.

Learning Objectives

After completing all phases, you should understand:

- Kubernetes Architecture
- Amazon EKS Fundamentals
- Pods and Deployments
- Services and Networking
- ConfigMaps and Secrets
- Persistent Storage
- AWS EBS Integration
- Ingress and External Access
- Horizontal Pod Autoscaling (HPA)
- Namespaces and Resource Governance
- Stateful Applications
- Monitoring with Prometheus and Grafana
- Infrastructure Cleanup and Cost Optimization

Phase 1 - EKS Foundation

Topics Covered:

- Amazon EKS Architecture
- Control Plane vs Worker Nodes
- kubectl Configuration
- Cluster Verification

Outcome: Successfully created and connected to an EKS cluster.

Phase 2 - Pods

Topics Covered:

- Pod Lifecycle
- Pod Scheduling
- Logs
- Pod Deletion
- Self-Healing Concepts

Outcome: Deployed and managed standalone pods.

Phase 3 - Deployments

Topics Covered:

- Deployments
- ReplicaSets
- Scaling
- Rolling Updates
- Self-Healing

Outcome: Managed applications using Deployments and multiple replicas.

Phase 4 - Services

Topics Covered:

- ClusterIP
- NodePort
- LoadBalancer

Outcome: Exposed applications internally and externally.

Phase 5 - ConfigMaps and Secrets

Topics Covered:

- Externalized Configuration
- Secret Management

- Environment Variables
- Volume Mounts

Outcome: Separated application configuration from container images.

Phase 6 - Persistent Storage

Topics Covered:

- Persistent Volumes (PV)
- Persistent Volume Claims (PVC)
- Storage Classes

Outcome:

Introduced persistent storage concepts in Kubernetes.

Phase 7 - AWS EBS Integration

Topics Covered:

- EBS CSI Driver
- Dynamic Volume Provisioning
- Persistent Data

Lab:

- Write data to a volume
- Delete pod
- Verify data survives

Outcome: Learned how Kubernetes integrates with AWS EBS storage.

Phase 8 - Ingress

Topics Covered:

- Ingress
- Ingress Controllers
- Path-Based Routing
- External Traffic Management

Outcome: Exposed applications through centralized routing.

Phase 9 - Horizontal Pod Autoscaling

Topics Covered:

- Metrics Server
- CPU Requests
- CPU Limits
- Horizontal Pod Autoscaler (HPA)

Lab:

- Generate CPU load
- Observe automatic scaling

Outcome: Implemented automatic workload scaling.

Phase 10 - Namespaces and Governance

Topics Covered:

- Namespaces
- ResourceQuota
- LimitRange

Outcome: Learned workload isolation and resource governance.

Phase 11 - Stateful Applications

Topics Covered:

- StatefulSets
- Headless Services
- Stable Pod Identity
- Persistent Storage per Pod

Lab:

- Stateful NGINX Deployment
- DNS-Based Pod Discovery
- Persistent EBS Volumes

Outcome:

Learned how stateful workloads differ from stateless workloads.

Phase 12 - Monitoring and Observability

Topics Covered:

- Helm
- Prometheus
- Grafana
- AlertManager
- Node Exporter
- Kube State Metrics

Lab:

- Deploy kube-prometheus-stack
- Visualize CPU and Memory Metrics
- Generate Load
- Observe Cluster Metrics

Outcome: Implemented a complete monitoring stack for EKS.

Phase 13 - Cleanup and Cost Optimization

Topics Covered:

- Resource Inventory
- Load Balancer Cleanup
- PVC Cleanup
- Monitoring Stack Removal
- Cluster Decommissioning

Outcome: Safely removed all AWS resources and minimized cloud costs.

Key Troubleshooting Scenarios Encountered

This repository also documents real issues encountered during hands-on practice:

- PVC Pending due to missing StorageClass
- StatefulSet immutable field errors
- Headless Service DNS troubleshooting
- HPA showing Unknown metrics
- Missing CPU Requests
- Label collisions between Deployments and StatefulSets

- Grafana access issues
- PVC stuck in Terminating state
- EKS cleanup and resource deletion verification

These troubleshooting exercises provided some of the most valuable learning throughout the journey.

Repository Structure

Typical files included:

```
deployment.yaml
service.yaml
configmap.yaml
secret.yaml
pv.yaml
pvc.yaml
ingress.yaml
hpa.yaml
headless-service.yaml
statefulset-storage.yaml
```

Important Notes

Never commit:

- AWS Credentials
- PEM Key Files
- kubeconfig Files
- Secrets containing real passwords

Always use:

- Kubernetes Secrets
- Environment Variables
- Secure Secret Management Practices

Final Outcome

This project demonstrates a complete Kubernetes lifecycle on Amazon EKS:

Cluster Creation

- Application Deployment
- Configuration Management
- Persistent Storage
- Networking
- Autoscaling
- Monitoring
- Troubleshooting
- Cleanup

The objective was not only to learn Kubernetes concepts but also to gain practical operational experience similar to real-world cloud environments.